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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 36: Material Flow Analysis (MFA)

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Concept of MFA: Mass Conservation

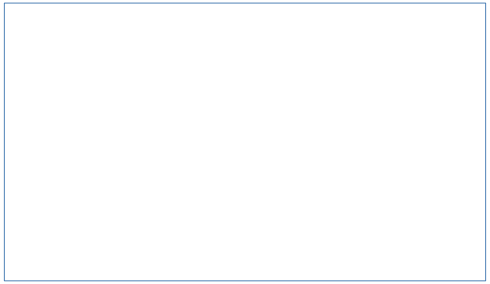
- MFA is based on the law of ‘Mass Conservation’
- Law of mass conservation: Matter can change form through physical and chemical changes, but through any of these changes, matter is conserved. The same amount of matter exists before and after the change—none is created or destroyed
- During any process, mass going into the system will be same as the sum of mass coming out of the system and mass absorbed in the system

The Law of Conservation of Mass dates from Antoine Lavoisier's 1789 discovery that mass is neither created nor destroyed in chemical reactions

Definition of Material Flow Analysis (MFA)

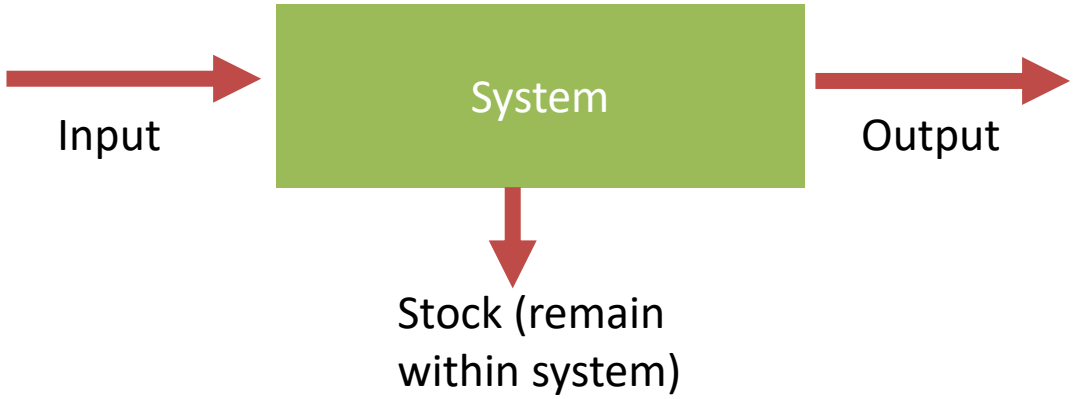
- Material flow analysis (MFA) is a systematic assessment of the flows and stocks of materials within a system defined in space and time.

  -Brunner and Rechberger, 2004



MFA concept equation

Input = Output + Stock



Example: If in any area 100 liters of fuel consumed with 2 gm Lead per liter and after combustion, 150 gm Lead is found in air then,
Input = 200 gm
Output = 150 gm.
Stock= Input- output= 200-150 =50 gm.

History of MFA

- The conservation of matter, or input equals output — was first postulated by Greek philosophers more than 2000 years ago.

- Long before MFA became a tool for managing resources, wastes, and the environment, the mass-balance principle has been applied in diverse fields as medicine, chemistry, economics, engineering, and life sciences.

- Two original areas of application were
  - Metabolism of cities
  - Analysis of pollutant pathways in regions such as watersheds or urban areas

Objectives/Purpose of MFA 1/2

- Delineate system of material flows and stocks
- Reduce system complexity while maintaining basis for decision-making
- Assess relevant flows and stocks quantitatively, checking mass balance, sensitivities, and uncertainties

Example: If any hazardous substance is causing health issues in any region, with the help of MFA we can create a flow and stock of that substance's incoming, outgoing and stocking in the region.

Objectives of MFA 2/2

- Present system results in reproducible, understandable, transparent fashion

- Use results as a basis for managing resources, the environment, and wastes
  - Monitor accumulation or depletion of stocks, future environmental loadings
  - Design of environmentally-beneficial goods, processes, and systems

Example: Vehicular emission of PM 10 is happening at large scale.

MFA can be useful to study how much environment loading is happening within a city region which further need to be addressed through mitigative measures.

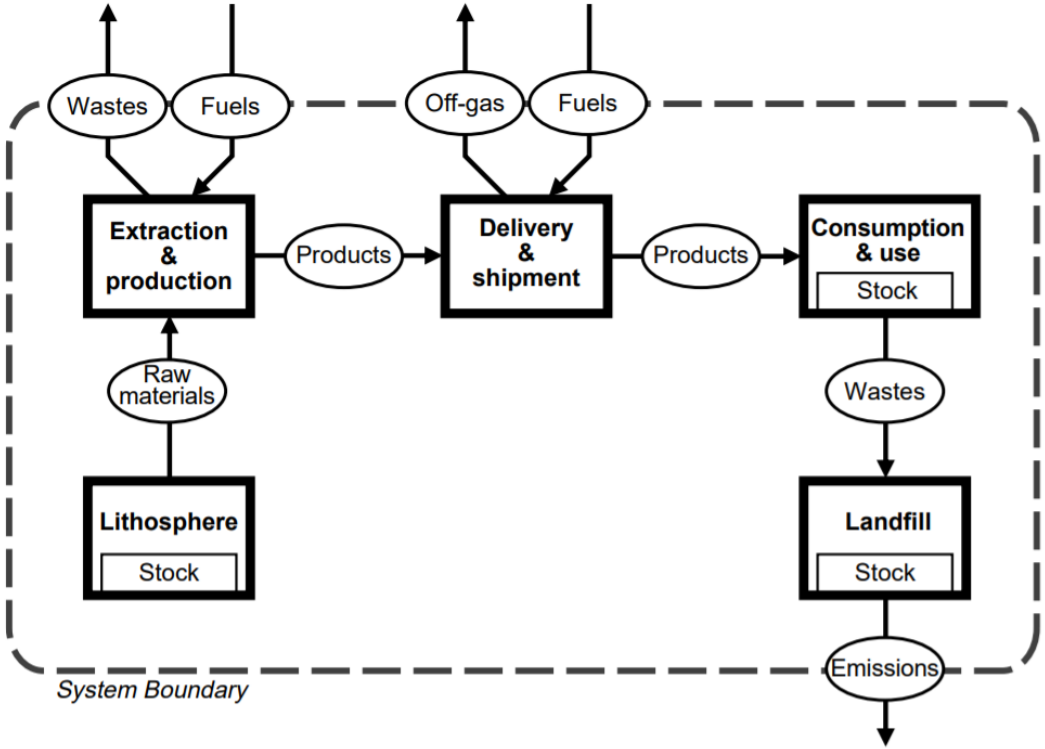
Important terms & definitions 1/3

- Material: substances and goods
- Substance: single type of matter (elements, compounds)
- Goods: substances or mixtures of substances that have economic values assigned by markets – Can include immaterial goods (energy, services, or information)
- Process: Transport, transformation, or storage of materials (natural or man-made)
  Example: Raw material to vehicle, construction of highway

Important terms & its definition 2/3

- System: Set of material flows, stocks, and processes within a defined boundary. Example: City boundary, region, Factory, household, etc.
- Activity: Set of systems needed to fulfill a basic human need (nourish, reside, transport, etc.)
- Stocks: Material reservoirs within the analyzed system

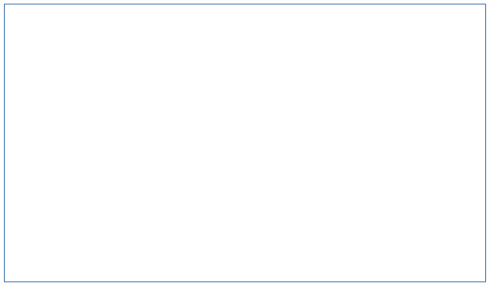
Example: Activity with system boundary



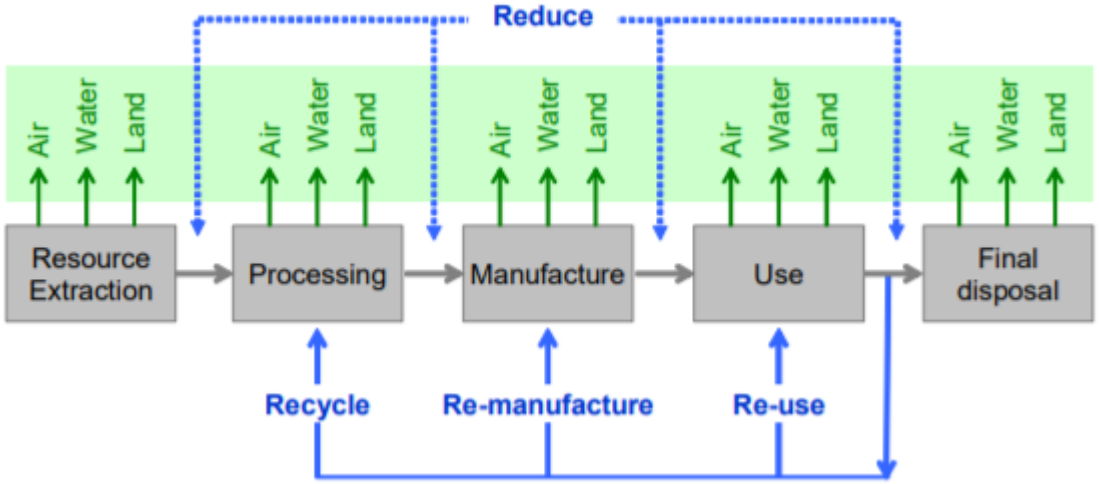
Various activities and interlinks are shown for a system for MFA study.

Important terms & definitions 3/3

- Flows: mass per time
- Fluxes: mass per time and cross section (per unit of area)
- Imports/exports: flows/fluxes across system boundaries
- Inputs/outputs: flows/fluxes across process boundaries

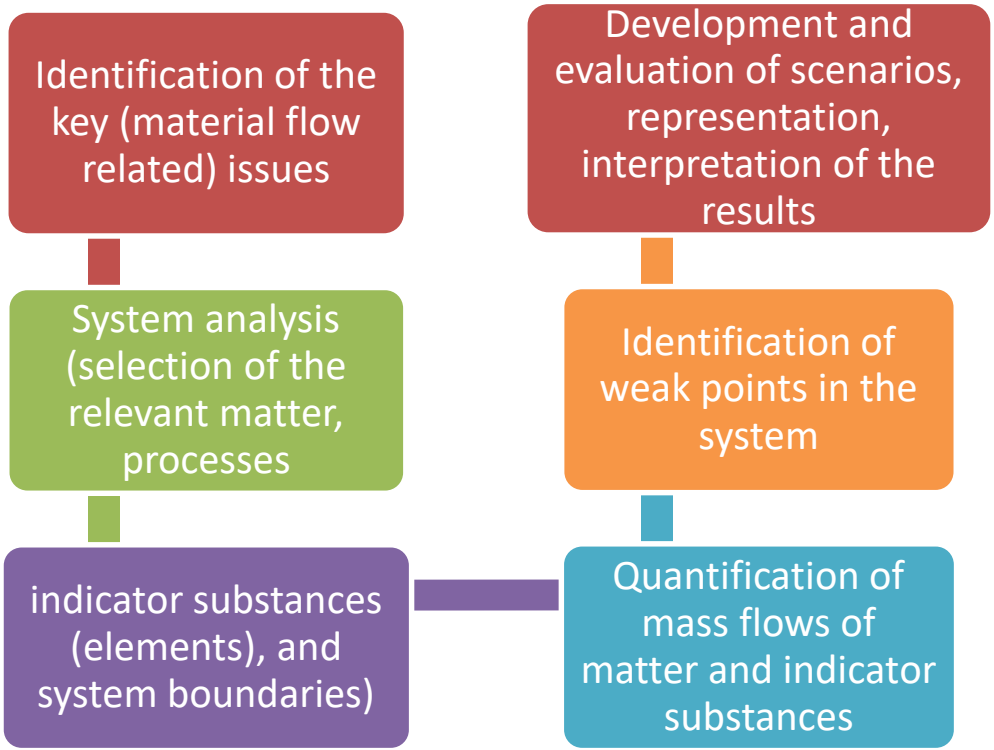


Flows of materials through the commercial life-cycle

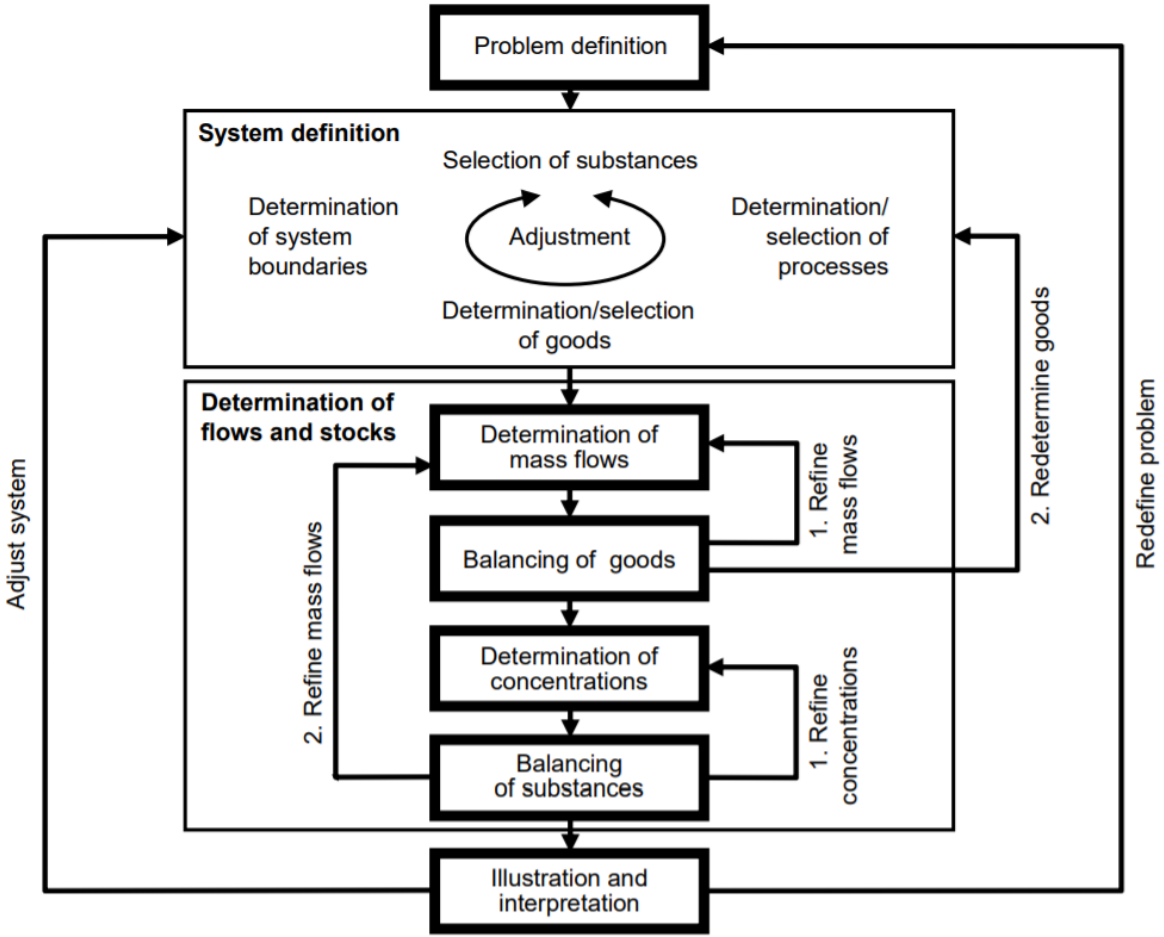


- Arrows represent process.
- Output is shown in green boxes.
- Through flow analysis, whole activity can be represented.

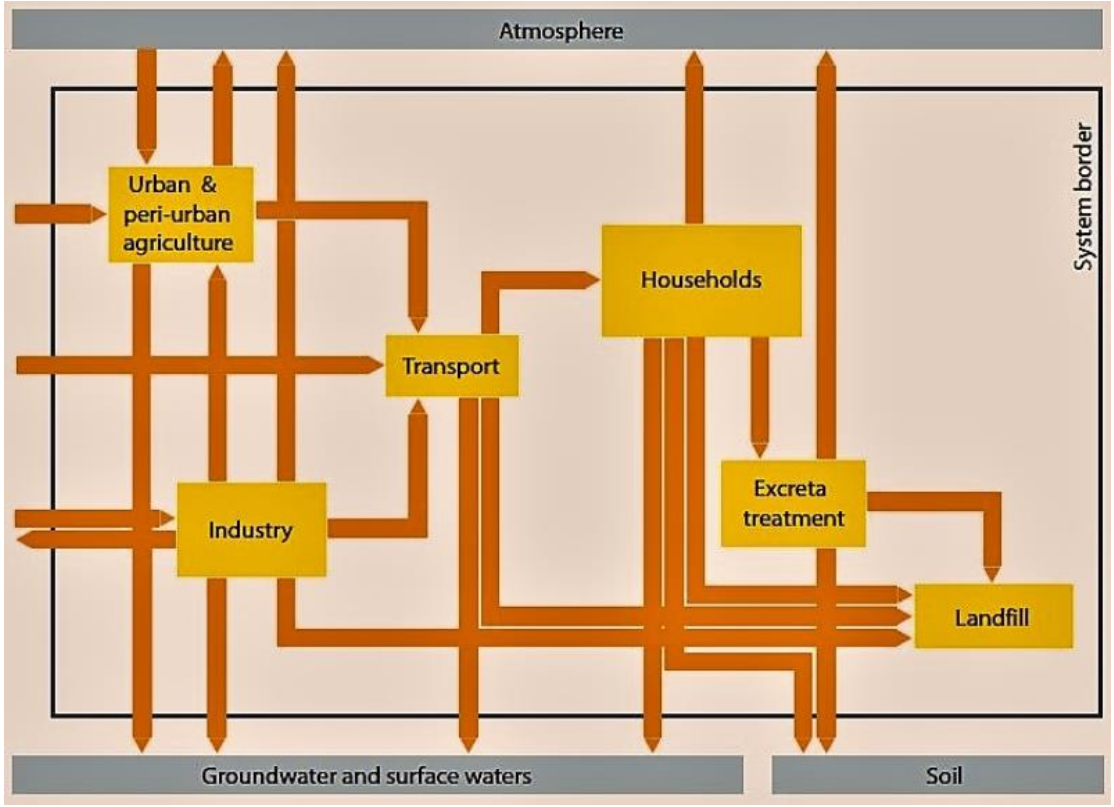
Procedure for MFA



MFA Process flow chart



Example: System



- Image showing a system, its boundaries, flows in a representative manner for waste generation.
- System diagram also depicts the connection between various activities within the system.

Source: FORSTER et al. (2003) and Lydie S.A. Yiougo & Dorothee Spuhler

Classification of MFA: Based on material in consideration

Parameter	Example
• Unit flow of material	Biomass, excavation…..
• Substance	Lead, Iron, CFC, CO_2….
• Product	Car, batteries…

Applications:

- Development of environmental policy for hazardous substances
- Evaluation of product's environmental impact

Classification of MFA: Based on system boundary

Parameter	Example
• Region	City, municipality etc.
• Sector	Transportation, construction….
• Firm/industry	Car manufacturing plant….

Application example:

- Providing firm environmental performance data
- Derivation of sustainability indicators
- Development of material flow accounts for use in official statistics

Application of MFA in environmental management

- Environmental impact statements
- Remediation of hazardous waste sites
- Design of air pollution control strategies
- Nutrient management in watersheds
- Planning of soil-monitoring systems

Ex- MFA can give clear picture of pollutant output from any thermal plant or region and accordingly mitigation strategies could be prepared.

Application of MFA in resource & waste management

- Analysis, planning and allocation, exploitation, and upgrading of resources

Ex.- Limiting the exploitation of natural resource, upgradation of battery for EVs etc.

- Modeling elemental compositions of wastes

Ex- What are the components of waste so that waste treatment can be done accordingly

- Evaluating material management performance in recycling/treatment facilities

Ex- Output of treatment facility should be lower than threshold concentration of any hazardous material

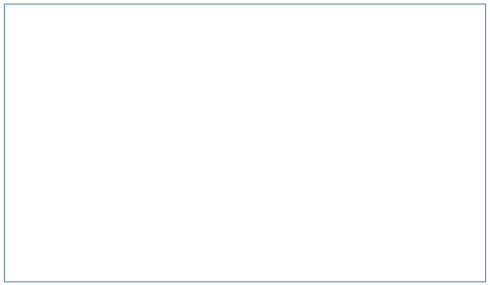
Application of MFA in Industrial ecology

- Controlling pathways for materials use and industrial processes
- Creating loop-closing industrial practices
- Dematerializing industrial output
- Systematizing patterns of energy use
- Balancing industrial input and output to natural ecosystem capacity

Ex- MFA-diagram should be analyzed, and change could be made in process so that resource saving procedure adopted.

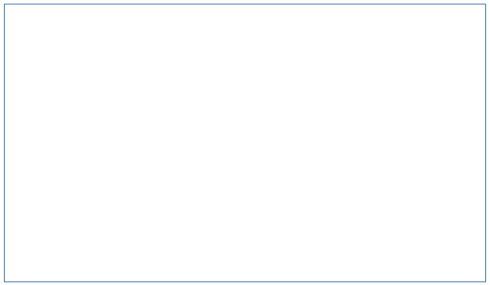
Advantages of MFA 1/2

- MFA allows having a critical view of resource management current status in a city
- MFA helps to evaluate the environmental soundness of various alternatives
- MFA can be used as a decision tool to choose sustainable technology



Advantages of MFA 2/2

- MFA is an ideal technical basis for planning and decision making, especially in developing and emerging countries with limited technical and financial resources
- In developing countries, MFA has been proven to be a suitable tool for early detection of environmental problems and development of appropriate solutions



Limitations of MFA

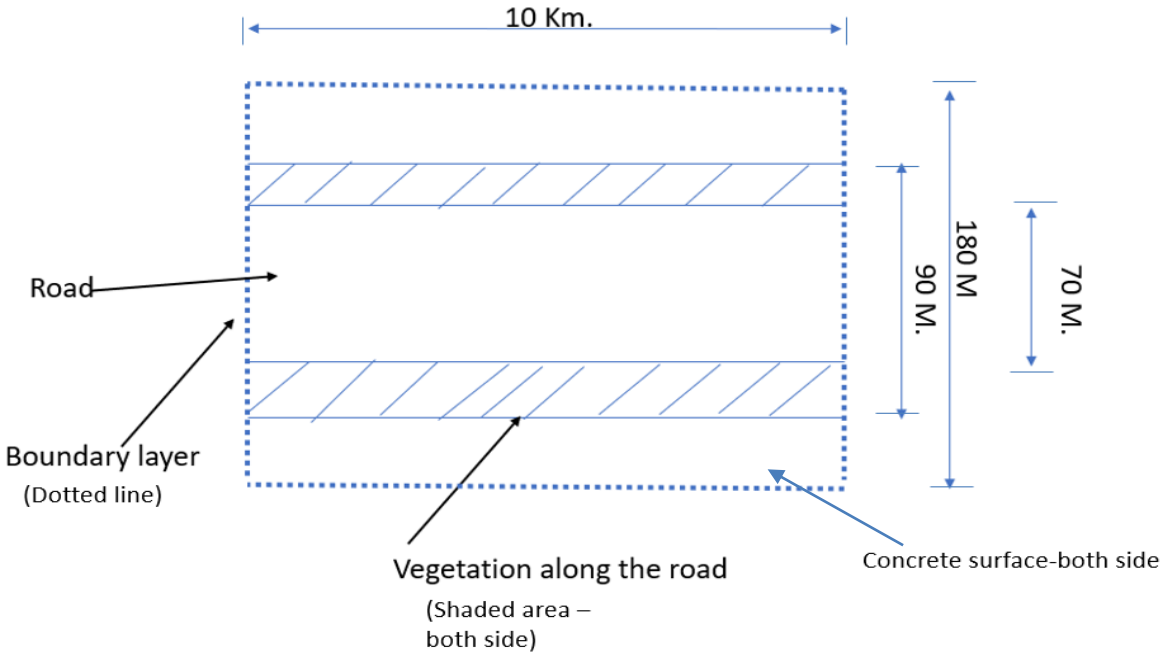
- Needs a lot of data to be implemented; there are only limited, reliable data available for developing countries
- There is a need to deal with uncertainties due to dynamic nature of various processes.

Numerical Example

A traffic survey has been conducted on a 10 km stretch of 2-way 2-lane road. Emission factors and traffic data are given in table. Shaded area in diagram is vegetation with absorbing capacity of 0.8 tonne CO_2 per 10000 square m per hour, while absorbing capacity of the road (carriage way) is 0.2 tonne CO_2 per 10000 square m per hour and rest of the surface (concrete surface) has 0.2 tonne CO_2 per 10000 square m per hour. Find input, output & stock in the system.

Vehicle Type	Emission (TonnCO2 per km)	Traffic in First Lane per hour	Traffic in Second Lane per hour
2 Wheelers	0.4	10	20
Cars	1	30	15
Small Trucks	2	15	20
Trucks	3	5	5
Bicycle	0.1	5	5
E-Cars	0.6	1	3

Numerical Example



Numerical Example

Stock = Absorption by surface of system

= Absorption capacity (ton/km^2/per hour) * area (km^2)*24 per day

= 0.2*concrete surface + 0.2* road area + 0.8* vegetation surface

= [0.2* (2*90) + 0.2* (70) + 0.8* (2*10)]*24

= 1584 ton

Input = CO_2 emission from vehicular movement within system

= (Emission from vehicle per km.*10) (*no. of vehicles/hr *24)

= 1640-ton CO_2 per day

Output = Input-stock

= 1640- 1584 = 56-ton CO_2/day

MFA, EIA vs LCA (1/2)

- LCA strives for completeness – As many substances as possible related to product considered for study.
- MFA strives for transparency and manageability – Limited number of substances considered during MFA.
- MFA can be a method to establish an inventory for an LCA.

Ex.: While doing LCA for a transportation project, MFA can be used in setting up the input/output material such as fuel, battery for EVs, further assessment of impacts will be done in LCA process.

MFA, EIA vs LCA (2/2)

- EIA is used for impact assessment of project.
- It generally considers impacts on local environment after the implementation of project.
- MFA on other hand assess the inflow & outflow of material for project area (system boundary). It doesn't analyse the impacts.

Ex.: For an airport project, MFA will study what materials are input for the projects and what are the outgoing materials whereas EIA will do assessment of impacts of activities due to construction and operation of airport. LCA can be done for any specific product such as aircrafts used in airport, or any material used during construction of the project.

Conclusions

- MFA assesses the flow of different material within the system.
- MFA can be used for various application such as environmental management, resource management & industrial ecology.
- Various alternatives can also be compared using MFA and thus most resource conservative option can be opted.

References

- Measuring material flows and resource productivity, Volume I., The OECD Guide; 2008

- Lydie S.A. Yiougo, Dorothee Spuhler “Material flow analysis”, SSWM University course module

- Paul H. Brunner and Helmut Rechberger; 2004, Practical Handbook of MATERIAL FLOW ANALYSIS

Thank You